T-Test

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Objectives

- Understand the concept of hypothesis testing.
- Learn how to apply a one-sample t-test.
- Use Python for hypothesis testing.

Hypothesis

- A **hypothesis** is just a guess or assumption about something.

Example:

You think **drinking coffee** before studying **improves test scores**.
This is your **hypothesis**—something you want to **test**.

Null Hypothesis

- H_o says “Nothing special is happening”. It assumes no effect or no difference.
- “Drinking coffee does not improve test scores”
- It means coffee has no real effect, and any difference in scores is just by chance.

Alternative Hypothesis

- H_1 says something is happening – There is a difference or effect.
- “Drinking coffee improves test scores”
- It means coffee actually makes a difference in performance.

What is a t-Test?

- A statistical test to compare a sample mean to a known value.

- Example: A fitness app claims users walk 10,000 steps per day on average. You collect data from 50 users and perform a t-test to see if the claim is accurate.

Why use t-test?

- Imagine you own a coffee shop.
- You claim that the average customer spends TK 500 per visit.
- To verify this claim, you collect data from 30 customers
- Perform a one-sample t-test to check if your claim holds true.

One-Sample T-Test

- Compare a sample to a known value
- Imagine you are in a **math class** where the teacher says that the **average exam score** for all students in the school is **70** out of 100.
- You want to **check if your class** (a sample of students) has an average score different from 70.
- To do this, you take the **scores of 10 students** from your class and perform a **one-sample t-test**.

Real life example

- Imagine your teacher says the average test score in the school is 70. (Assumption on population)
- You test 10 students from your class to see if your class is different.
- **H_0 (Null Hypothesis):** Your class has the same average score as the school [**sample mean equals the population mean ($\mu=\mu_0$).]**

- **H_1 (Alternative Hypothesis):** Your class has a different average score than your school (not 70) [**The sample mean does not equal the population mean ($\mu\neq\mu_0$) for a two-tailed test.]**

T-Test Formula

t = t-statistic (measures how far the sample mean is from the population mean in terms of standard error)

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Where:

- $\bar{x}$: Sample mean
- μ_0: Hypothesized population mean
- s: Sample standard deviation
- n: Sample size

Steps to Perform a One-Sample t-Test in Python

- Claim: The teacher says the average exam score **in the school** is 70
- Step 1: Collect Data

  Let's assume these are the exam scores of 10 students **in your class**:

  [65, 72, 68, 75, 80, 62, 71, 78, 74, 69]. You want to check if the average score is **significantly different** from 70 using one-sample t-test
- Step 2: Define Hypotheses
  - Null Hypothesis (H_0): The average class score is equal to 70.
  - Alternative Hypothesis (H_1): The average class score is not equal to 70.

Significance Level (α)

- The probability of rejecting the null hypothesis when it is actually true.
- It represents the risk of making Type I error (False Positive)
- α = 0.05 (Means we accept a 5% chance of wrongly concluding that the class's average is not 70 when it is actually 70 (equal to population's mean)

P-value

- The p-value tells us the probability of getting a result this far from 70, just by chance.
- If the p-value is less than α, reject H_0.
- Lower p-value means stronger evidence against H_0
- Higher p-value means not enough evidence to reject H_0

Degrees of Freedom

- The number of data points that are free to vary
- In one-sample t-test, usually, df = n-1, where n is sample size
- You need mean to calculate, so you assume mean should not get the freedom to vary (assuming mean is in data point)

Assumptions

- Data is continuous.
- Data follows a normal distribution.
- Sample observations are independent.